

STROTHER FIELD, KS, USA

WMO: 724502

Lat: 37.168N

Lon: 97.037W

Elev: 1150

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 13932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	14.4	-0.4	5.6	13.5	3.4	6.8	17.0	29.6	43.6	26.8	44.7	8.3	0	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	100.2	74.5	97.3	74.8	93.7	74.6	78.5	91.7	77.5	90.7	76.6	89.2	13.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	137.3	85.7	73.5	130.1	83.8	72.7	126.7	83.1	43.0	91.6	41.7	90.4	40.7	89.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.6	24.2	20.9	1.0	103.8	5.6	3.5	-3.0	106.3	-6.3	108.4	-9.5	110.3	-13.6	112.8		
			0.2	80.6	5.4	1.2	-3.7	81.5	-6.9	82.2	-9.9	82.9	-13.8	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	35.3	39.1	48.8	58.1	67.7	77.1	81.7	80.4	72.9	60.2	48.1	37.0
	DBStd	18.68	10.50	11.40	10.74	9.25	7.91	5.93	5.55	5.88	7.63	9.37	9.97	10.14
	HDD50	1572	467	334	151	32	1	0	0	0	0	22	149	416
	HDD65	4072	922	726	509	243	67	2	0	0	19	205	511	868
	CDD50	4840	10	28	114	275	550	812	982	942	686	337	92	12
	CDD65	1865	0	0	7	35	151	364	517	477	255	55	4	0
	CDH74	20425	1	3	66	332	1327	3896	6268	5501	2513	485	33	0
	CDH80	9676	0	0	9	71	435	1765	3316	2848	1095	133	4	0
(14) Wind	WSAvg	9.9	9.7	10.3	11.4	11.9	10.1	10.3	9.1	8.3	8.8	9.6	10.0	9.6
(15) Precipitation	PrecAvg	37.5	1.1	1.4	3.0	3.7	5.1	6.0	4.0	3.9	2.9	3.1	2.0	1.4
	PrecMax	53.9	2.4	3.6	7.4	9.3	13.2	15.2	8.2	7.2	6.3	9.9	6.1	3.6
	PrecMin	26.0	0.0	0.0	0.5	0.6	2.2	0.9	0.7	0.3	0.9	0.3	0.0	0.2
	PrecStd	7.4	0.8	1.0	1.9	1.9	2.5	3.5	1.7	1.9	1.7	2.3	1.9	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.6	73.0	80.4	86.5	92.5	100.0	104.3	104.4	100.1	88.8	78.5	68.0
		MCWB	51.8	56.7	61.9	67.0	71.1	74.3	74.3	74.1	72.9	69.0	61.4	56.8
	2%	DB	62.6	67.8	74.9	81.5	88.3	95.8	100.3	100.1	93.2	83.7	72.9	62.9
		MCWB	49.9	53.8	60.2	64.7	71.5	74.7	75.1	74.2	73.0	66.6	59.6	53.2
	5%	DB	57.3	62.6	70.5	78.0	84.5	92.1	97.4	97.2	90.4	79.4	68.5	57.1
		MCWB	46.5	51.5	57.5	62.8	70.2	74.5	75.5	74.4	72.6	65.2	57.9	48.8
	10%	DB	51.9	57.0	66.3	73.5	81.6	89.6	93.6	93.0	86.7	75.1	63.8	53.0
		MCWB	43.1	47.8	55.5	61.5	69.6	73.9	75.5	74.4	71.2	64.1	55.0	46.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	60.5	67.0	70.1	76.3	79.8	80.6	80.0	77.3	73.6	66.2	61.8
		MCDB	61.8	67.1	75.2	81.3	86.6	92.9	92.6	92.7	90.5	82.2	73.1	65.9
	2%	WB	51.0	57.0	63.5	67.9	74.5	78.0	78.6	78.2	75.5	71.3	62.3	55.0
		MCDB	59.6	64.7	70.3	77.0	83.8	90.4	92.0	91.5	87.6	78.4	69.1	59.2
	5%	WB	47.0	53.3	60.6	66.0	72.7	76.7	77.6	77.0	74.1	68.9	59.5	50.2
		MCDB	55.9	60.7	68.0	74.6	81.5	88.2	91.2	90.1	85.9	75.5	66.1	56.2
	10%	WB	43.6	48.3	56.9	63.6	70.8	75.1	76.7	75.6	72.8	65.8	56.6	45.9
		MCDB	51.5	57.1	65.1	72.1	79.2	85.7	89.9	88.0	83.8	73.1	63.4	51.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	21.5	22.3	22.0	20.4	20.1	20.4	21.0	21.6	22.0	21.7	19.5
		MCDBR	29.9	29.1	27.9	26.5	23.2	22.9	24.6	26.1	24.1	25.4	26.1	26.5
	5% WB	MCWBR	18.7	18.3	16.4	14.7	10.9	8.6	7.5	7.9	9.2	12.5	15.6	17.8
		MCWBR	27.1	26.1	22.7	21.8	19.7	20.0	20.3	21.0	20.7	19.6	21.8	22.3
(40) Clear-Sky Solar Irradiance	taub		0.293	0.301	0.340	0.378	0.414	0.419	0.429	0.430	0.386	0.342	0.303	0.284
		taud	2.488	2.476	2.395	2.314	2.285	2.302	2.315	2.254	2.365	2.475	2.517	2.526
	Ebn at Noon		286	298	293	286	276	274	270	267	274	278	280	283
		Edh at Noon	25	29	35	40	42	41	41	42	36	29	25	23
(44) All-Sky Solar Radiation	RadAvg		813	1035	1347	1688	1877	2146	2124	1909	1593	1185	874	693
	RadStd		82	152	88	114	174	170	127	111	123	138	95	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page